

the Jupyter notebook

- 1 the Jupyter notebook
 - documenting your work
 - the four major operations
 - formatting with restructured text
- 2 Exact, Symbolic, and Approximate Computations
 - three types of computations
 - using the working precision

MCS 320 Lecture 2
Introduction to Symbolic Computation
Jan Verschelde, 15 June 2026

the Jupyter notebook

1 the Jupyter notebook

- documenting your work
- the four major operations
- formatting with restructured text

2 Exact, Symbolic, and Approximate Computations

- three types of computations
- using the working precision

the Jupyter notebook

The Jupyter notebook is a web based interface for computations.

Jupyter stands for Julia, Python, R, and many others.

- Input cells can contain code or text (markdown, $\text{\LaTeX}$).
- Output cells hold numerical data, symbolic expressions, or plots.
- We use Jupyter to document our computations.

Jupyter is the default interface for SageMath.

A well structured notebook runs as a computer program.

All your work must be submitted in a Jupyter notebook.

the Jupyter project is separate from SageMath



T. Kluyver, B. Ragan-Kelley, F. Pérez, B. Granger, M. Bussonnier, J. Frederic, K. Kelley, J. Hamrick, J. Grout, S. Corlay, P. Ivanov, D. Avila, S. Abdalla, C. Willing, and Jupyter Development Team: **Jupyter Notebooks — a publishing format for reproducible computational workflows.**

In F. Loizides and B. Schmidt, editors, *Positioning and Power in Academic Publishing: Players, Agents, and Agendas*, pages 87–90. IOS Press, 2016.

Visit <https://jupyter.org> for more information.

the Jupyter notebook

1 the Jupyter notebook

- documenting your work
- the four major operations
- formatting with restructured text

2 Exact, Symbolic, and Approximate Computations

- three types of computations
- using the working precision

the four major operations

Making, downloading, uploading, and printing:

- 1 When making a Jupyter notebook, we select a kernel.
- 2 Download (or saving) a Jupyter notebook to your disk.
Be aware that many browsers will add the `.txt` extension to the `.ipynb` file.
- 3 Upload a notebook into a SageMath session.
The end of the `.ipynb` file contains meta information, about the kernel and the version of the notebook.
- 4 Printing to a pdf file.
This operation is useful to generate a technical report.

the Jupyter notebook

1 the Jupyter notebook

- documenting your work
- the four major operations
- **formatting with restructured text**

2 Exact, Symbolic, and Approximate Computations

- three types of computations
- using the working precision

formatting with restructured text

The Markdown cells occur in two types:

- 1 headers: section (#), subsection (##), subsubsection (###),
- 2 paragraphs of text.

To edit a formatted Markdown cell, convert to `Raw NBConvert`.

To learn more, visit

<https://docutils.sourceforge.io/rst.html>

the Jupyter notebook

1 the Jupyter notebook

- documenting your work
- the four major operations
- formatting with restructured text

2 Exact, Symbolic, and Approximate Computations

- three types of computations
- using the working precision

Exact, Symbolic, and Approximate Computations

We distinguish between three types of computations:

- 1 An *exact* computation is free from errors.

Example: $\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$.

- 2 A *symbolic* computation operates on symbols.

Example: $\sin(\pi/3) = \sqrt{3}/2$.

Note $\sqrt{3}$ is the symbol for the unevaluated function `sqrt(3)`.

- 3 An *approximate* computation works in limited precision.

Example: $\sin(3.1415/3) = 0.8660$.

the Jupyter notebook

1 the Jupyter notebook

- documenting your work
- the four major operations
- formatting with restructured text

2 Exact, Symbolic, and Approximate Computations

- three types of computations
- using the working precision

Using the Working Precision

Consider the following computation: `Pi10 = pi.n(digits=10)`

The `Pi10` evaluates to `3.141592654`, an approximation for π , accurate with 10 decimal places.

The statement `delta = pi - Pi10` shows `pi - 3.141592654`, which is a symbolic expression.

Now consider the following two statements:

- 1 `delta.n(digits=10)` evaluates to `0.0000000000`
- 2 `delta.n(digits=11)` evaluates to `3.6379788071e-12`

Which outcome is correct? Explain.